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Duran, Carlxen Brieyl P. COM232 - ML Exercise 5 - Gaussian Mixture Models

Person	Height	Weight	r_w	r_m
A	158	52	0.98	0.02
B	162	56	0.95	0.05
C	166	60		
D	175	72	0.10	0.90
E	180	78	0.05	0.95
F	185	84	0.02	0.98

Initial Parameters	Women	Men
mean (μ)	$\begin{bmatrix} 160 \\ 55 \end{bmatrix}$	$\begin{bmatrix} 180 \\ 78 \end{bmatrix}$
mixing coefficient (π)	0.5	0.5
covariance (Σ)	$\begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix}$	$\begin{bmatrix} 16 & 0 \\ 0 & 25 \end{bmatrix}$

1. Compute the responsibility for sample C

$$\mathcal{N}(C \mid \mu_w, \Sigma_w) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_w|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(C-\mu_w)^T \Sigma_w^{-1} (C-\mu_w)\right)$$

WOMEN

$$\mu_w = \begin{bmatrix} 160 \\ 55 \end{bmatrix} \quad \Sigma_w = \begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix} \quad d = 2$$

$$(2\pi)^{\frac{d}{2}} |\Sigma_w|^{\frac{1}{2}} = |\Sigma_w| = 9 \times 16 = 144 \quad = |\Sigma_w|^{\frac{1}{2}} = 144^{\frac{1}{2}} = 12$$

$$= (2\pi)^{\frac{d}{2}} = 2\pi = 6.283$$

$$\frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_w|^{\frac{1}{2}}} = \frac{1}{6.283 \times 12} = \frac{1}{75.40}$$

$$(C-\mu_w)^T \Sigma_w^{-1} (C-\mu_w) = \frac{(x_h - \mu_h)^2}{\cancel{10}} + \frac{(x_w - \mu_w)^2}{\cancel{10}} = \frac{(166-160)^2}{9} + \frac{(60-55)^2}{16}$$

$$= \frac{36}{9} + \frac{25}{16} = 4 + 1.5625 = 5.5625$$

$$\mathcal{N}(C \mid \mu_w, \Sigma_w) = \frac{1}{75.40} \exp\left(-\frac{1}{2} \times 5.5625\right) = \frac{1}{75.40}(0.062)$$

$$= \frac{1}{75.40} \exp(-2.78125) = 8.22 \times 10^{-4}$$

MEN

$$\mu_m = \begin{bmatrix} 180 \\ 78 \end{bmatrix} \quad \Sigma_m = \begin{bmatrix} 16 & 0 \\ 0 & 25 \end{bmatrix} \quad d = 2$$

$$(2\pi)^{\frac{d}{2}} |\Sigma_m|^{\frac{1}{2}} = |\Sigma_m| = 16 \times 25 = 400 \quad = |\Sigma_m|^{\frac{1}{2}} = 400^{\frac{1}{2}} = 20$$

$$= (2\pi)^{\frac{d}{2}} = 2\pi = 6.283$$

$$\frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_m|^{\frac{1}{2}}} = \frac{1}{6.283 \times 20} = \frac{1}{125.66}$$

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$$(c-\mu_m)^T \Sigma_m^{-1} (c-\mu_m) = \frac{(x_h-\mu_h)^2}{10} + \frac{(x_m-\mu_m)^2}{10} = \frac{(166-180)^2}{16} + \frac{(60-78)^2}{25}$$

$$= \frac{196}{16} + \frac{324}{25} = 12.25 + 12.96 = 25.21$$

$$\mathcal{N}(c \mid \mu_m, \Sigma_m) = \frac{1}{125.66} \exp\left(-\frac{1}{2} \times 25.21\right) = \frac{1}{125.66}\left(3.34 \times 10^{-6}\right)$$

$$= \frac{1}{125.66} \exp(-12.605) = 2.66 \times 10^{-8}$$

Responsibilities $\quad r_{c,k} = \dfrac{\pi_k \mathcal{N}(c \mid \mu_k, \Sigma_k)}{\Sigma_j \pi_j \mathcal{N}(c \mid \mu_j, \Sigma_j)}$

$\mathcal{N}_w = 8.22 \times 10^{-4} \qquad \mathcal{N}_m = 2.66 \times 10^{-8} \qquad \pi_w = \pi_m = 0.5$

WOMEN

$$\frac{8.22 \times 10^{-4}}{8.22 \times 10^{-4} + 2.66 \times 10^{-8}} \boxed{= 0.99997}$$

MEN

$$\frac{2.66 \times 10^{-8}}{8.22 \times 10^{-4} + 2.66 \times 10^{-8}} \boxed{= 0.00003}$$

(2.) Compute the data points for each cluster

$m_c = \sum_{i=1} r_{ic} \qquad m_{women} = 3$

$\qquad\qquad\qquad m:_{men} = 3$

(3.) Compute the updated mixing coefficient of each cluster

$\pi_{women} = \frac{3}{6} = \frac{1}{6} \boxed{= 0.5} \qquad \pi_{men} = \frac{3}{6} = \frac{1}{2} \boxed{= 0.5} \qquad \pi_c = \frac{m_c}{m}$

(4.) Compute the updated means for each cluster

WOMEN $\quad \mu_c = \frac{1}{m_c} \sum r_{ic} x^i \qquad$ MEN $\quad \mu_c = \frac{1}{m_c} \sum r_{ic} x^i$

$\mu_{height} = \frac{158 + 162 + 166}{3} = 162 \qquad \mu_{height} = \frac{175 + 180 + 185}{3} = 180$

$\mu_{weight} = \frac{52 + 56 + 60}{3} = 56 \qquad \mu_{weight} = \frac{72 + 78 + 84}{3} = 78$

$\boxed{\mu_{women} = \begin{bmatrix} 162 \\ 56 \end{bmatrix}} \qquad \boxed{\mu_{men} = \begin{bmatrix} 180 \\ 78 \end{bmatrix}}$

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(5.) Compute the updated covariance of each cluster $\Sigma_c = \frac{1}{m_c} \sum_{n=i} r_{ic} (X_i - \mu_c)^T (X_i - \mu_c)$

WOMEN

$\mu_{women} = \begin{bmatrix} 162 \\ 56 \end{bmatrix}$ $A = \begin{bmatrix} 158 \\ 52 \end{bmatrix}$ $B = \begin{bmatrix} 162 \\ 56 \end{bmatrix}$ $C = \begin{bmatrix} 166 \\ 60 \end{bmatrix}$

$(A - \mu_{women}) \left(\begin{bmatrix} 158 \\ 52 \end{bmatrix} - \begin{bmatrix} 162 \\ 56 \end{bmatrix} \right) = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$ $(A - \mu_{women})^T = [-4, -4]$

$(A - \mu_{women})(A - \mu_{women})^T = \begin{bmatrix} -4 \\ -4 \end{bmatrix} [-4, -4] = \begin{bmatrix} 16 & 16 \\ 16 & 16 \end{bmatrix}$

$(B - \mu_{women}) \left(\begin{bmatrix} 162 \\ 56 \end{bmatrix} - \begin{bmatrix} 162 \\ 56 \end{bmatrix} \right) = [0]$

$(C - \mu_{women}) \left(\begin{bmatrix} 166 \\ 60 \end{bmatrix} - \begin{bmatrix} 162 \\ 56 \end{bmatrix} \right) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ $(C - \mu_{women})^T = [4, 4]$

$(C - \mu_{women})(C - \mu_{women})^T = \begin{bmatrix} 4 \\ 4 \end{bmatrix} [4, 4] = \begin{bmatrix} 16 & 16 \\ 16 & 16 \end{bmatrix}$

$\Sigma_{women} = \frac{1}{3} \begin{bmatrix} 16 & 16 \\ 16 & 16 \end{bmatrix} + [0] + \begin{bmatrix} 16 & 16 \\ 16 & 16 \end{bmatrix}$

$\Sigma_{women} = \frac{1}{3} \begin{bmatrix} 32 & 32 \\ 32 & 32 \end{bmatrix}$

$\boxed{\Sigma_{women} = \begin{bmatrix} 10.67 & 10.67 \\ 10.67 & 10.67 \end{bmatrix}}$

MEN

$\mu_{men} = \begin{bmatrix} 180 \\ 78 \end{bmatrix}$ $D = \begin{bmatrix} 175 \\ 72 \end{bmatrix}$ $E = \begin{bmatrix} 180 \\ 78 \end{bmatrix}$ $F = \begin{bmatrix} 185 \\ 84 \end{bmatrix}$

$(D - \mu_{men}) = \left(\begin{bmatrix} 175 \\ 72 \end{bmatrix} - \begin{bmatrix} 180 \\ 78 \end{bmatrix} \right) = \begin{bmatrix} -5 \\ -6 \end{bmatrix}$ $(D - \mu_{men})^T = [-5, -6]$

$(D - \mu_{men})(D - \mu_{men})^T = \begin{bmatrix} -5 \\ -6 \end{bmatrix} [-5, -6] = \begin{bmatrix} 25 & 30 \\ 30 & 25 \end{bmatrix}$

$(E - \mu_{men}) = \left(\begin{bmatrix} 180 \\ 78 \end{bmatrix} - \begin{bmatrix} 180 \\ 78 \end{bmatrix} \right) = [0]$

$(F - \mu_{men}) = \left(\begin{bmatrix} 185 \\ 84 \end{bmatrix} - \begin{bmatrix} 180 \\ 78 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ $(F - \mu_{men})^T = [5, 6]$

$(F - \mu_{men})(F - \mu_{men})^T = \begin{bmatrix} 5 \\ 6 \end{bmatrix} [5, 6] = \begin{bmatrix} 25, & 30 \\ 30 & 25 \end{bmatrix}$

$\Sigma_{men} = \frac{1}{3} \begin{bmatrix} 25 & 30 \\ 30 & 25 \end{bmatrix} + [0] + \begin{bmatrix} 25 & 30 \\ 30 & 25 \end{bmatrix}$

$\Sigma_{men} = \frac{1}{3} \begin{bmatrix} 50 & 60 \\ 60 & 50 \end{bmatrix}$

$\boxed{\Sigma_{men} = \begin{bmatrix} 16.67 & 20 \\ 20 & 16.67 \end{bmatrix}}$